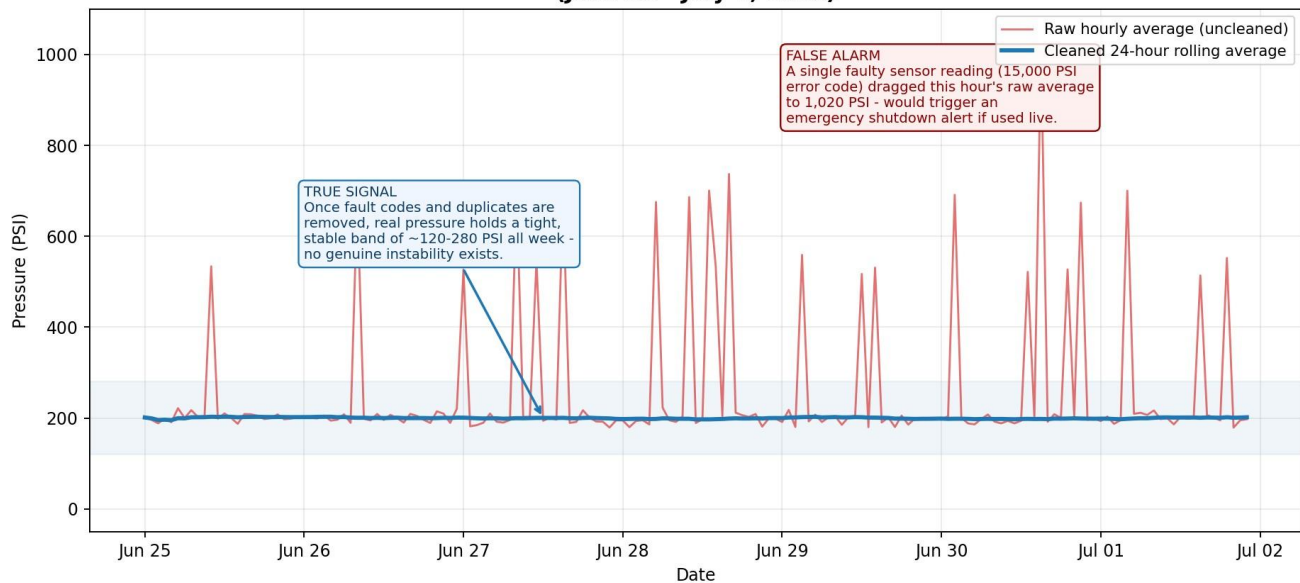


Week 2 Insight Report

Plant Pressure Monitoring: Raw Feed vs. Cleaned Signal (June 25 - July 1, 2026)



CONTEXT: What was wrong with the raw data?

The raw sensor feed for the week of June 25–July 1 is not fit for direct use. Out of 5,015 logged readings, 99 pressure and temperature values were equipment fault codes impossible readings like 15,000 PSI, -273°C, and a repeated exact-zero dropout code mixed in alongside real measurements with no flag to tell them apart. On top of that, zone names were logged in 15 different spellings for only 5 real locations, 15 rows were duplicated, and 124 readings were missing outright. The practical effect: a single fault code was enough to drag an entire hour's average pressure to over 1,020 PSI on the raw chart above (red line) a level that would trip an emergency shutdown alert if this feed were connected to a live monitoring dashboard today.

INSIGHT: What did cleaning actually reveal?

Once the fault codes, duplicates, and inconsistent labels were removed, the real picture is the opposite of alarming: pressure holds a stable, tight band of roughly 120–280 PSI across the entire week (blue line above), with no meaningful difference between Morning, Afternoon, or Night shifts (198–202 PSI average across all three) and no single Zone running dangerously hot or over-pressured. We also checked whether the faulty readings clustered around a specific shift or time of day, in case this pointed to an operational stress pattern they did not. Fault codes are spread evenly across all three shifts and all hours of the day, which points to an intermittent sensor or transmission fault, not a process problem.

ACTION: What should change operationally?

First, do not trust raw pressure alerts from this feed until it is fixed the current system can generate false emergency signals, which risks alarm fatigue among floor staff. Second, have the instrumentation team inspect the sensors and transmission link responsible for the fault codes; because the errors are evenly spread rather than tied to one zone, this looks like a shared logging/transmission issue rather than a single failing unit. Third, add an automatic fault-code filter (rejecting readings outside 100–400 PSI and 10–150°C) directly on the live ingestion pipeline, not just in after-the-fact reporting, so dashboards and alerts reflect the true 120–280 PSI operating band going forward.